Educational research prototype. This application demonstrates computational analysis of DNA mutations and simplified machine-learning approaches to mutation-impact prediction. It does not diagnose genetic diseases, determine clinical pathogenicity, replace genetic testing, provide medical advice, or substitute for professional genetic interpretation.

1	163	110	24	92.5%	400
Mutations analyzed (this session)	Missense (demo) · Demo Dataset	Synonymous (demo) · Demo Dataset	Nonsense (demo) · Demo Dataset	Best-model accuracy (demo) · Demo Dataset	Dataset size (demo) · Demo Dataset

Workflow

DNA → Mutation detection → Codon analysis → Amino-acid translation → Mutation classification → ML impact prediction → Explanation

REFINED RESEARCH QUESTION

Can AI predict the potential impact of a DNA mutation?

How effectively can computational sequence analysis and supervised machine learning classify coding DNA mutations and estimate their potential functional impact using features derived from nucleotide, codon, amino-acid and variant information?

How mutations affect proteins

DNA → mRNA → Codons (triplets) → Amino acids → Protein

DNA uses four bases: A (adenine), T (thymine), G (guanine), C (cytosine). During protein synthesis, bases are read in groups of three called codons. A mutation can leave the amino acid unchanged (**synonymous**), change it (**missense**), create a premature stop codon (**nonsense**), or add/remove bases (**insertion/deletion**). If the number of bases added or removed is not a multiple of three, the downstream reading frame shifts (**frameshift**). Consequences depend on position, gene context and protein structure, which this prototype does not model.

Mutation Analyzer

ORIGINAL DNA (A/T/G/C ONLY)

ATGGAATTTCCG

MUTATED DNA

ATGGCATTTCCG

READING FRAME (FIXED START POINT — FIRST CODON BEGINS AT BASE…)

1 (default)

The reading frame is a fixed offset you choose, not something derived from the sequence's length. It stays the same for both sequences below; codons are read as non-overlapping triplets from that one starting point. A trailing stretch shorter than 3 bases simply can't form a full codon — that's a length/truncation fact, not a "shift" in frame.

Analysis result

Sequences (changed bases highlighted; | separates codons in frame 1)

```
Original  ATG | GAA | TTT | CCG
Mutated   ATG | GCA | TTT | CCG
```

1 variant site detected. Each is classified separately: 1 missense.

#	POSITION	BASE CHANGE	CODON #	CODON CHANGE	AMINO ACID	CONSEQUENCE
1	5	A → C	2	GAA → GCA	Glu → Ala	MISSENSE

Protein sequence changed: YES

Protein Change Viewer (translation halts at the first stop codon; highlighted = differs)

Original: Met **Glu** Phe Pro

Mutated: Met **Ala** Phe Pro

Explanation (generated from the computed result)

Variant 1 (position 5, A → C): GAA → GCA changes Glutamic acid to Alanine without a stop codon, so this variant is missense. Whether it affects function depends on context this tool does not model.

Codon Translator

DNA CODON

GAA

DNA GAA → mRNA GAA

Glutamic acid (Glu / E)

Genetic code (standard)

TTT Phe	TTC Phe	TTA Leu	TTG Leu	TCT Ser	TCC Ser	TCA Ser	TCG Ser	TAT Tyr	TAC Tyr	TAA Stop	TAG Stop	TGT Cys
TGC Cys	TGA Stop	TGG Trp	CTT Leu	CTC Leu	CTA Leu	CTG Leu	CCT Pro	CCC Pro	CCA Pro	CCG Pro	CAT His	CAC His
CAA Gln	CAG Gln	CGT Arg	CGC Arg	CGA Arg	CGG Arg	ATT Ile	ATC Ile	ATA Ile	ATG Met	ACT Thr	ACC Thr	ACA Thr
ACG Thr	AAT Asn	AAC Asn	AAA Lys	AAG Lys	AGT Ser	AGC Ser	AGA Arg	AGG Arg	GTT Val	GTC Val	GTA Val	GTG Val
GCT Ala	GCC Ala	GCA Ala	GCG Ala	GAT Asp	GAC Asp	GAA Glu	GAG Glu	GGT Gly	GGC Gly	GGA Gly	GGG Gly	

Protein Change Viewer

Example (ATGGAATTTCCG → ATGGCATTTCCG, reading frame 1). Each amino acid is a chip; differences are highlighted.

Original: Met **Glu** Phe Pro

Mutated: Met **Ala** Phe Pro

Dataset Explorer

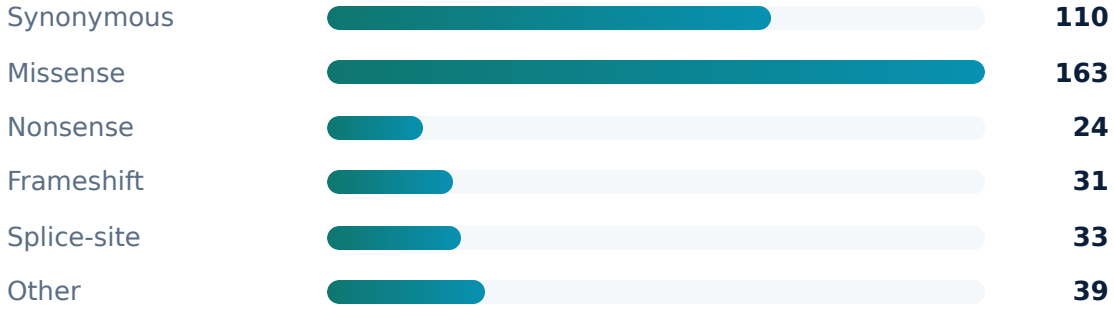
Educational Demonstration Dataset. 400 synthetic variants generated in your browser (seeded, seed 42). The impact_class label is produced by a hand-written scoring rule plus noise. It is **not** real biological data, not from ClinVar, and gene names are placeholders. "Higher impact" here does not mean disease-causing.

400 of 400 variants match. Showing first 10.

VARIANT_ID ▲	GENE	VARIANT_TYPE	MOLECULAR_CONSEQUENCE	ALLELE_FREQUENCY	PROTEIN_IMPACT	VARIANT_LENGTH	IMPACT_CLASS
DEMO-0001	GENE_B	SNV	Missense	0.031	Low	1	0
DEMO-0002	GENE_C	SNV	Missense	0.0122	Low	1	0
DEMO-0003	GENE_B	SNV	Synonymous	0.0207	Low	1	0
DEMO-0004	GENE_C	SNV	Missense	0.0114	Moderate	1	0
DEMO-0005	GENE_B	SNV	Splice-site	0.0104	Moderate	1	1
DEMO-0006	GENE_A	SNV	Synonymous	0.0241	Unknown	1	0
DEMO-0007	GENE_B	SNV	Synonymous	0.0058	Low	1	0
DEMO-0008	GENE_D	SNV	Missense	0	Low	1	0
DEMO-0009	GENE_E	SNV	Synonymous	0.0005	Unknown	1	0
DEMO-0010	GENE_B	SNV	Splice-site	0.0004	High	1	1

Dataset Explorer (2/2)

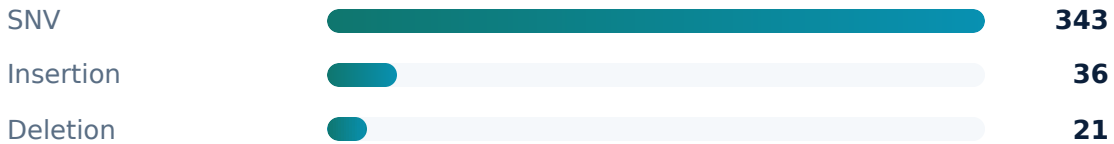
Consequence distribution



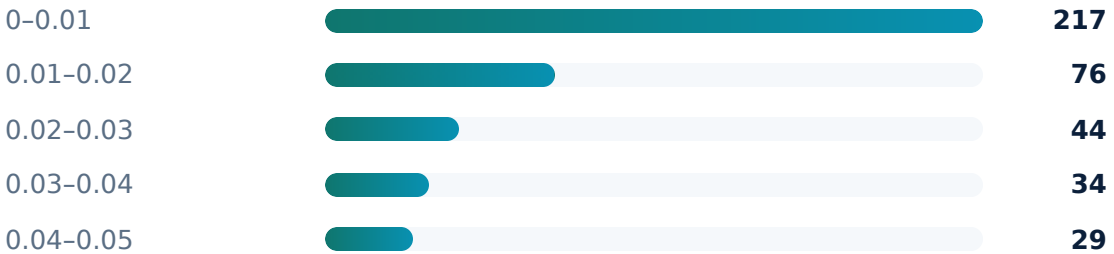
Impact class distribution



Variant-type distribution



Allele frequency (bins)



Mutation Impact Predictor

Features are encoded and fed to the models trained on the demonstration dataset. Use the Analyzer first, then copy its result here, or enter features manually.

GENE (TEXT; NOT USED BY THE MODEL)

GENE_DEMO

VARIANT TYPE

SNV

MOLECULAR CONSEQUENCE

Missense

PROTEIN IMPACT

Moderate

ALLELE FREQUENCY (0–1)

0.001

VARIANT LENGTH (BASES)

1

MODEL

Logistic Regression

Protein Stability & Interaction Impact (SDM / mCSM / DUET — simplified)

At the protein level, a missense mutation's consequence depends on how the new residue fits the local 3D environment: whether it still packs into a buried core, keeps the right size and chemistry, and preserves contacts with other residues, ligands or partner molecules. Three structure-based methods developed in the Blundell lab address this: **SDM** is a statistical method that scores a substitution using environment-specific substitution tables built from thousands of homologous protein structures; **mCSM** represents the atomic neighbourhood around the mutated residue as a graph and uses machine learning trained on experimental stability/affinity data to score the change; **DUET** combines both into one consensus prediction. The real tools are published and freely usable: SDM2, mCSM, DUET.

> **This calculator is a simplified, original re-implementation for teaching purposes — not the real SDM/mCSM/DUET.** The real SDM uses 216 substitution tables derived from 2,054 structural alignments (12,038 structures); the real mCSM/DUET are machine-learning models trained on the ProTherm and SKEMPI thermodynamic databases. Here, the *same log-odds formula shape* SDM uses is kept, but the substitution probabilities are approximated from Grantham's (1974) physicochemical distance (composition, polarity, volume) instead of trained structural tables, and the mCSM-style graph signature is approximated from user-entered neighbour counts instead of real 3D coordinates. Treat every output as illustrative only — it will not match the real servers.

Model Performance (1/2)

Results from Educational Demonstration Dataset. These numbers do not represent clinical genetic prediction accuracy. The data are synthetic and the label is rule-derived, so high scores mostly show the models can recover the rule.

Metrics on held-out test set (80 variants)

MODEL	ACCURACY	PRECISION	RECALL	F1	ROC-AUC
Logistic Regression	0.925	1.000	0.684	0.813	0.976
Random Forest	0.925	1.000	0.684	0.813	0.995

Confusion matrix — Logistic Regression

	pred 0	pred 1
true 0	61	0
true 1	6	13

Confusion matrix — Random Forest

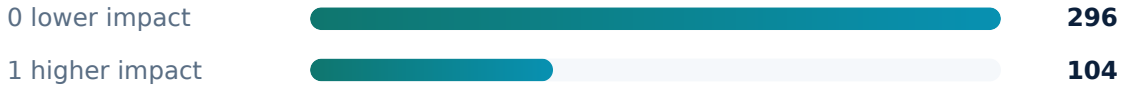
	pred 0	pred 1
true 0	61	0
true 1	6	13

Model Performance (2/2)

Model comparison (F1 / AUC)



Class distribution



Feature importance in the mutation impact model



Random Forest impurity-based importance, grouped per original feature. It shows how useful a feature was for this trained model; it does not prove that the feature biologically causes disease or determines clinical pathogenicity. Gene name is excluded because it does not generalise as a predictor.

Pipeline

Generate/validate → Select features → One-hot + standardise (train only) → 80/20 split → Train LR + RF (n=100, seed 42) → Evaluate

Research & Limitations (1/3)

Research Process

1. **Broad topic:** how genetic mutations affect proteins.
2. **Application:** AI-based mutation impact prediction.
3. **Five sub-problems:** mutation detection → codon analysis → mutation classification → impact prediction → evaluation.
4. **Selected:** can machine learning estimate mutation impact from variant features?
5. **Clarifying answers:** a mutation is a DNA change; not every mutation changes a protein; "impact" is a simplified model estimate, not clinical pathogenicity; the app cannot say whether a person has a disease.
6. **Bioinformatics:** DNA → codon → amino acid → mutation type (rule-based, standard genetic code).
7. **Machine learning:** features → Logistic Regression (batch gradient descent, fixed learning rate, 800 epochs, L2-regularized — this browser build stands in for scikit-learn's LogisticRegression) and Random Forest (100 trees, seed 42), compared on a held-out 20% test set with preprocessing fit on training data only.
8. **Evaluation:** accuracy, precision, recall, F1, ROC-AUC, confusion matrix.

Research & Limitations (2/3)

Research questions

1. What are DNA mutations and how do they occur?
2. How do codons relate to amino acids?
3. How can substitutions alter protein sequences?
4. How do synonymous, missense and nonsense mutations differ?
5. How do insertions and deletions affect coding sequences?
6. How can mutation information become computational features?
7. Can ML estimate a simplified impact category?
8. Which model performs better on the selected dataset?
9. What are the limitations?

Conclusion

This project demonstrates how a biological question about genetic mutations can be transformed into a computational bioinformatics and machine-learning problem. DNA sequences can be compared to identify mutations, affected codons and amino-acid changes, and to classify selected consequences. Machine-learning models can explore patterns between variant-level features and simplified impact categories. However, a computational prediction does not by itself establish whether a mutation causes disease; these tools support exploring genetic variation and do not replace experiments or clinical genetic interpretation.

References (general sources; no dataset records used)

- NCBI ClinVar — future real-data source
- NCBI Bookshelf
- National Human Genome Research Institute
- Ensembl
- UniProt
- PubMed
- scikit-learn documentation

Links point to site home pages; add specific articles you have actually read and verified.